



Universidad Gerardo Barrios

Cede: Usuluán

Facultad: Ciencia y tecnología

Materia: Estadística Computacional

Catedrático: ING. Alcir Gustavo Quintanilla Rodríguez

TEMA: Laboratorio 1 del computo 3

Alumnos: Jaime Arnulfo Segovia González – USSS018724

Olimpia Esmeralda Rivas Vanegas – USSS017124

FECHA: 12/11/25

Ejercicio ① Alumnos: Jaime Amulfo Segovia González

Datos

Olimpia Esmeralda Rivas Vanegas

2, 2, 3, 4, 4, 6, 7, 7, 8, 8, 9, 9, 9, 10, 10, 10, 11, 12
 13, 13, 14, 15, 15, 16, 17, 18, 19, 20

$$n = 28$$

$$\frac{K \cdot n}{m}$$

Encuentre

a) P_{25} y q_0 b) D_5 c) Q_3

$$* P_{25} = \frac{(25)(28)}{100} = 7.0$$

$$P_{25} = 7$$

$$P_{25} = \frac{\text{valor 7} + \text{valor 8}}{2}$$

$$P_{25} = \frac{7 + 7}{2} = \frac{14}{2} = 7$$

$$* P_{90} = \frac{(90)(28)}{100} = 25.2$$

$$P_{90} = \text{valor } (25 + 1)$$

$$P_{90} = \text{valor } 26$$

$$P_{90} = 18$$

$$* Q_3 = \frac{(3)(28)}{4} = 21.0$$

$$Q_3 = \frac{\text{valor 21} + \text{valor 22}}{2}$$

$$Q_3 = \frac{14 + 15}{2} = 14.5$$

$$Q_3 = 14.5$$

$$* D_5 = \frac{(5)(28)}{10} = 14.0$$

$$D_5 = \frac{\text{valor 14} + \text{valor 15}}{2}$$

$$D_5 = \frac{10 + 10}{2} = \frac{20}{2} = 10$$

$$D_5 = 10$$

Ejercicio 2

Respuesta del Ejercicio 2

Int.	f _a	f _{rel}
1-7	4	4
8-14	10	14
15-21	7	21
22-28	6	27
29-35	3	30

Total 30

$$l_i = 8, f_a = 4, F_i = 10, i_c = 7$$

$$P_{25} = \frac{25(30)}{100} = 7.5$$

$$P_{25} = 8 + \left(\frac{7.5 - 4}{10} \right) (7) = 9.95 \text{ días}$$

$$P_{40} = \frac{40(30)}{100} = 12$$

$$l_i = 22, f_a = 21, F_i = 6, i_c = 7$$

$$P_{40} = 22 + \left(\frac{12 - 21}{6} \right) (7) = 28.5$$

b) Decil

$$D_5 = \frac{5(30)}{10} = 15$$

$$l_i = 14.5, f_a = 14, F_i = 7, i_c = 7$$

$$D_5 = 14.5 + \left(\frac{15 - 14}{7} \right) (7) = 15.5$$

Tercer Cuartil

$$Q_3 = \frac{3(30)}{4} = 22.5$$

$$l_i = 21, f_u = 21, f_i = 6, c = 7$$

$$Q_3 = 21.5 + \frac{(22.5 - 21) \cdot 7}{6} = 23.25$$

$$Q_3 = 23.25 \text{ días}$$

$$D_5 = 15.5 \text{ días}$$

$$P_{25} = 9.95 \text{ días}$$

$$P_{90} = 28.5 \text{ días}$$